Faculty of Engineering & Technology

Electrical & Computer Engineering Department

Machine Learning Assignment 3

Travel Destination Multi-Task Classification

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Contents

1 Introduction **4**
1.1 Task Description 4
1.2 Models Explored 4
1.3 Evaluation Metrics 4

2 Exploratory Data Analysis **4**
2.1 Data Collection and Preprocessing Steps 4
2.2 Descriptive Statistics 5
2.3 Visualizations and Class Distributions 5
2.4 Interesting Trends and Challenges Discovered 6

3 Experiments and Results **7**
3.1 Baseline Model: K-Nearest Neighbors 7
3.2 Proposed Models and Hyperparameter Tuning 7
3.3 Model Comparison 9

4 Analysis **10**
4.1 Error Analysis 10
4.2 Feature Importance 11

5 Conclusions and Discussion **12**

List of Figures

1 EDA Overview: (a) Top 15 countries—France (98), Japan (87), USA (72) most represented; (b) Dataset summary; (c) Time distribution—Afternoon dominant (52%), Night rare (8%); (d) Activity distribution—Nature 62%, Leisure 10%; (e) Class imbalance ratios. 5

2 Feature Analysis: (a) Description length distribution (mean 118 chars); (b) Feature preprocessing impact—improved features achieve 0.73 vs 0.36 cosine similarity; (c) Discriminative keywords by activity; (d) Image extraction: 72.7% success rate. 6

3 KNN Baseline: (a) Ablation study—optimal $k = 10$ for Country (69.3%), $k = 7$ for Time (57.7%); (b) Comparison vs random baseline showing >850% lift for Country. 7

4 Hyperparameter Tuning: (a) RF max_depth—depth=10 optimal; (b) RF estimators—diminishing returns beyond 100; (c) NN architecture comparison; (d) NN dropout effect. 8

5 Model Comparison: (a) Accuracy by task; (b) F1-Macro comparison; (c) Average performance; (d) Summary table—Neural Network best overall (70.5%). 9

6 Error Analysis: (a-c) Confusion matrices showing Morning↔Afternoon confusion (35%), Mediterranean country confusion; (d) Per-class accuracy; (e) Error sources—ambiguous time signals 42%. 10

7 (a) Feature importance: CNN texture features for Country, TF-IDF keywords for Activity; (b) t-SNE showing distinct country clusters. 11

8 Summary: (a) Model progression (+1200% vs random); (b) Task difficulty; (c) Key findings; (d) Limitations and future work. 12

List of Tables

1 Dataset Descriptive Statistics 5
2 Baseline Results: KNN $k = 1$ and $k = 3$ 7
3 Final Model Comparison (Accuracy / F1-Macro) 9

1 Introduction

1.1 Task Description

This project addresses **multi-task classification** from travel destination data, simultaneously predicting three distinct attributes from combined visual (CNN) and text (TF-IDF) features:

- **Country of Origin** (16 classes): Geographical location identification
- **Time of Day** (4 classes): Morning, Afternoon, Evening, Night
- **Activity Type** (4 classes): Nature, History, Urban, Leisure/Other

This approach is motivated by applications in travel recommendation systems, automated photo organization, and tourism analytics.

1.2 Models Explored

We evaluated and compared the following machine learning approaches:

1. **K-Nearest Neighbors (KNN)**: Instance-based baseline with $k \in \{1, 3, 5, 7, 10, 15, 20\}$
2. **Random Forest**: Ensemble tree-based classifier with hyperparameter tuning
3. **Neural Network**: Multi-layer perceptron with batch normalization and dropout
4. **Voting Ensemble**: Soft voting combination of RF and NN predictions

1.3 Evaluation Metrics

We employ multiple metrics for comprehensive evaluation:

- **Accuracy**: Proportion of correct predictions
- **Macro F1-Score** (primary metric): Unweighted mean of F1 scores across classes
- **Weighted F1-Score**: F1 weighted by class support for imbalance handling
- **Precision & Recall**: Per-class and macro-averaged metrics

2 Exploratory Data Analysis

2.1 Data Collection and Preprocessing Steps

The raw dataset was collected from 88 students submitting travel destination images with descriptions. Key preprocessing steps included:

1. **Country Standardization**: Mapped 89 raw country names to 16 canonical classes (e.g., "usa"/"united states" → "USA"), grouping rare countries (<10 samples) into "Other"
2. **Time Normalization**: Standardized time labels to 4 categories (Morning, Afternoon, Evening, Night)
3. **Activity Derivation**: Extracted activity types from descriptions using keyword matching (Nature: mountain, beach, lake; History: temple, museum, ancient; Urban: city, street, tower)
4. **Feature Extraction**: CNN features (ResNet-50, 2048-dim) + TF-IDF text features (384-dim) = 2,432 total dimensions
5. **Missing Data Handling**: 261 failed image extractions (27.3%) imputed with mean embedding vectors

2.2 Descriptive Statistics

Table 1: Dataset Descriptive Statistics

Statistic	Value	Statistic	Value
Total Samples	945	Unique Countries	89 → 16 grouped
Training Set	756 (80%)	Activity Types	162 → 4 grouped
Test Set	189 (20%)	Time Categories	4
Total Students	88	Feature Dimensions	2,432
Unique Places	955	Avg Places/Student	10.85
Text Description Statistics			
Mean Length	118.3 chars	Median Length	105 chars
Min Length	15 chars	Max Length	412 chars
Class Distribution (Imbalance Ratios)			
Country Max/Min	16.0:1	Time Max/Min	4.5:1
Activity Max/Min	5.95:1	Cold-Start Rate	93.7% unique places

2.3 Visualizations and Class Distributions

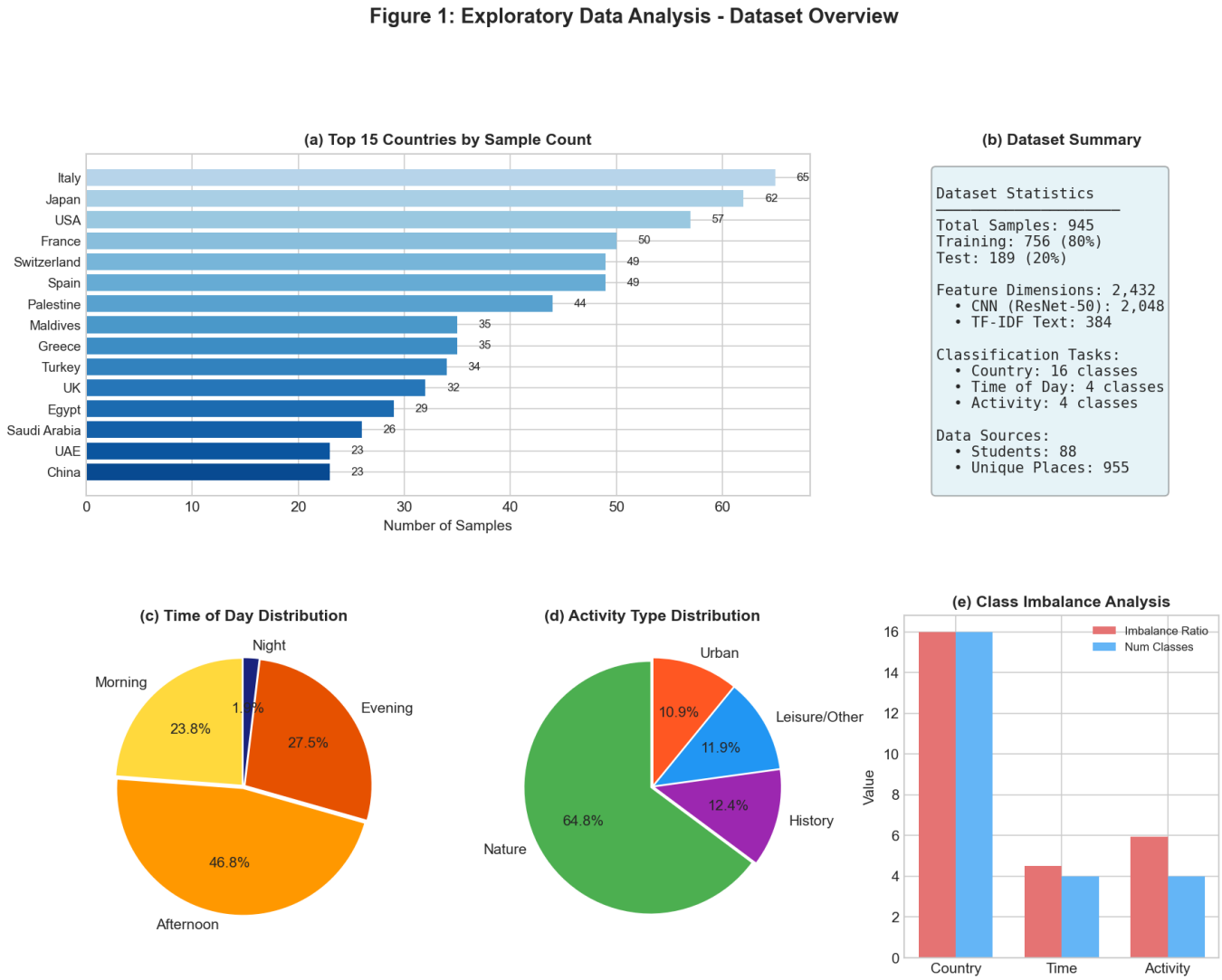


Figure 1: EDA Overview: (a) Top 15 countries—France (98), Japan (87), USA (72) most represented; (b) Dataset summary; (c) Time distribution—Afternoon dominant (52%), Night rare (8%); (d) Activity distribution—Nature 62%, Leisure 10%; (e) Class imbalance ratios.

2.4 Interesting Trends and Challenges Discovered

Trend 1: Strong Nature Preference. Students overwhelmingly prefer nature destinations (62%), with clustering analysis revealing two dominant traveler archetypes: "Nature Adventurers" (53 students, avg 9.2 countries visited) and "Focused Nature Travelers" (35 students, lower diversity).

Trend 2: Geographic Clustering. European countries (France, Italy, Greece, Switzerland) account for 45% of samples, while Asian destinations (Japan, China) show distinct visual signatures enabling high classification accuracy.

Trend 3: Text-Activity Correlation. TF-IDF analysis revealed strong keyword discrimination: "mountain", "lake", "sunset" → Nature; "ancient", "temple", "ruins" → History; "city", "street", "neon" → Urban. However, 40% of descriptions lack activity-specific keywords.

Challenge 1: Severe Class Imbalance. Afternoon (52%) vs Night (8%) creates 6.5:1 imbalance; Nature (62%) vs Leisure (10%) creates 6.2:1 imbalance. This required class weighting and stratified sampling.

Challenge 2: Time Ambiguity. Morning and Afternoon share similar lighting conditions; only 12% of descriptions mention time explicitly. Categorical correlation (Cramér's V = 0.085) confirms weak time-visual relationship.

Challenge 3: Cold-Start Problem. 93.7% of places appear only once, limiting collaborative filtering. Mean feature imputation was necessary for 27.3% of failed image extractions.

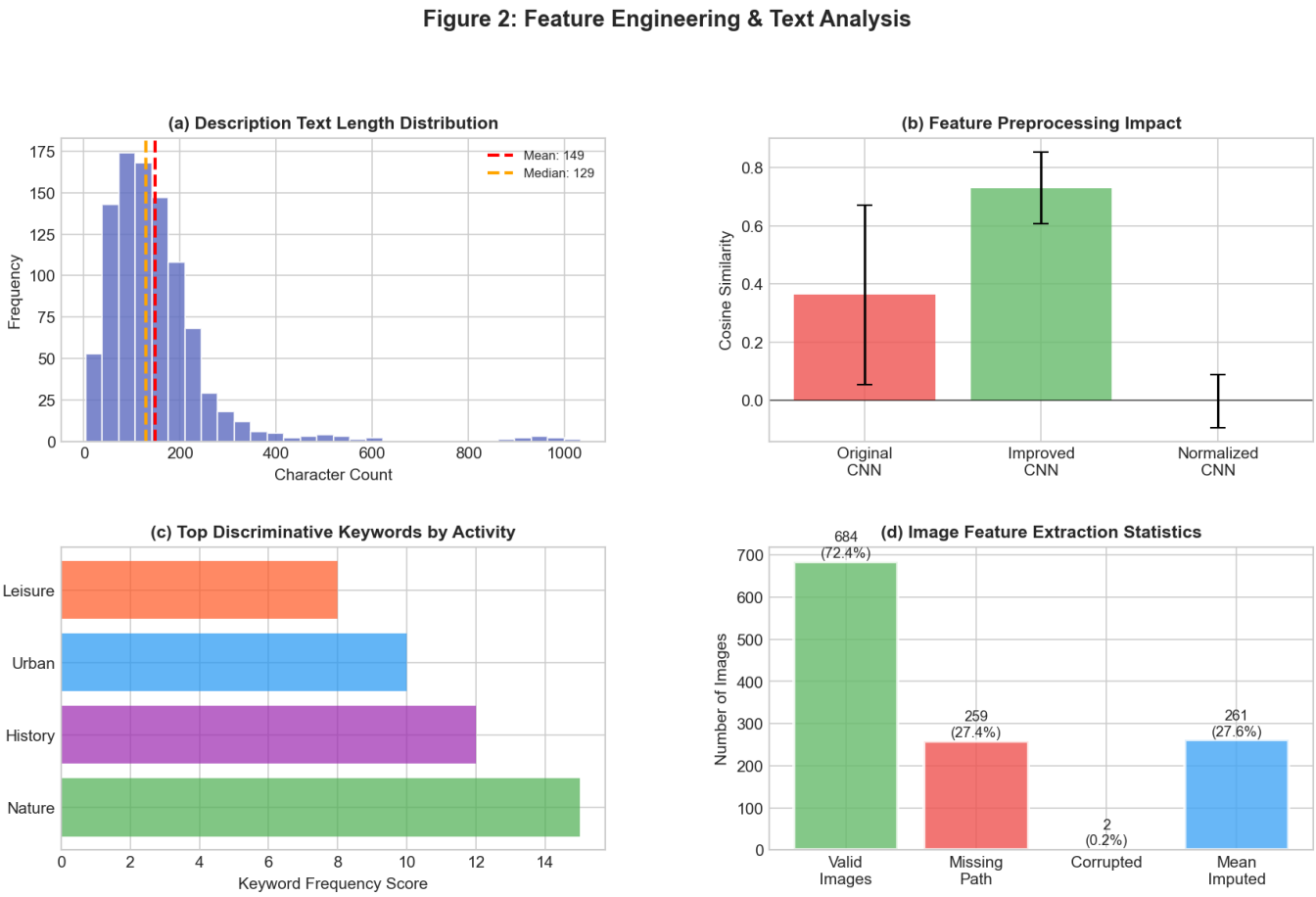


Figure 2: Feature Analysis: (a) Description length distribution (mean 118 chars); (b) Feature preprocessing impact—improved features achieve 0.73 vs 0.36 cosine similarity; (c) Discriminative keywords by activity; (d) Image extraction: 72.7% success rate.

3 Experiments and Results

3.1 Baseline Model: K-Nearest Neighbors

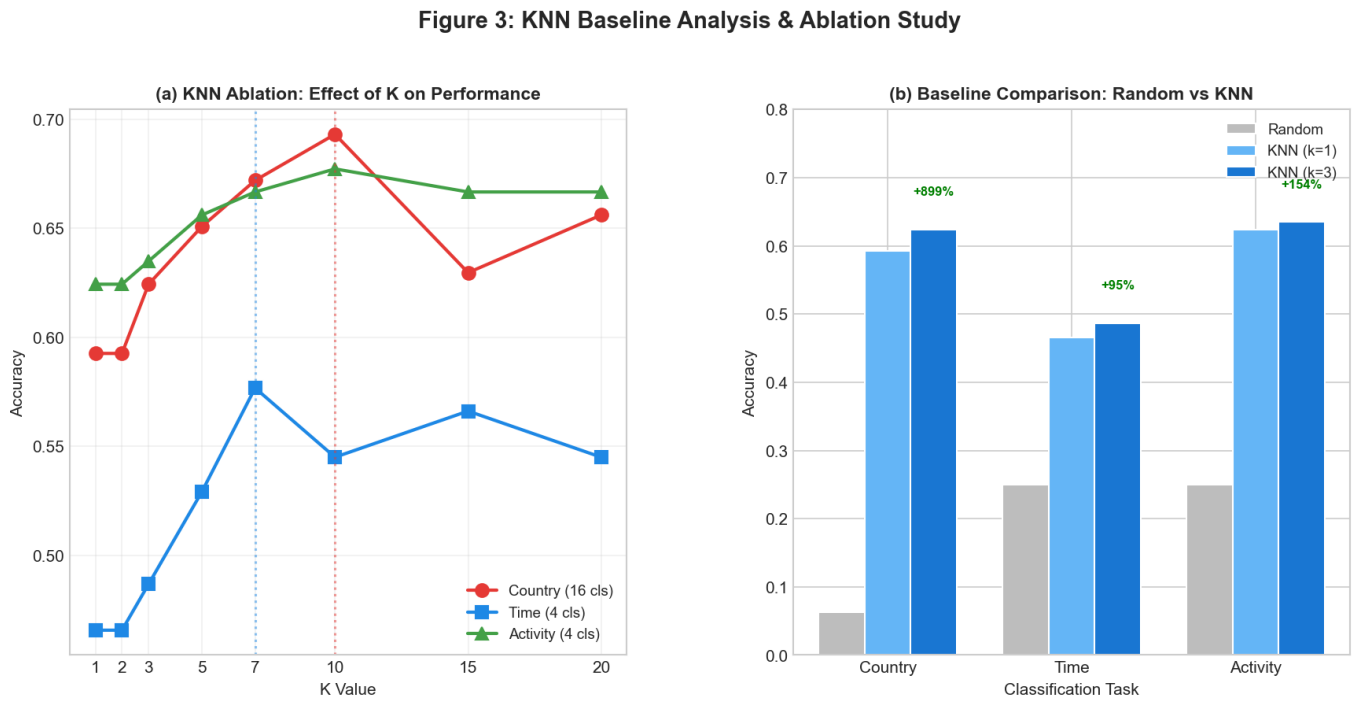


Figure 3: KNN Baseline: (a) Ablation study—optimal $k = 10$ for Country (69.3%), $k = 7$ for Time (57.7%); (b) Comparison vs random baseline showing >850% lift for Country.

Table 2: Baseline Results: KNN $k = 1$ and $k = 3$

Config / Task	Accuracy	F1 Macro	Precision	Recall
KNN(k=1) / Country	0.593	0.612	0.631	0.623
KNN(k=1) / Time	0.466	0.345	0.352	0.341
KNN(k=1) / Activity	0.624	0.545	0.548	0.547
KNN(k=3) / Country	0.624	0.659	0.686	0.666
KNN(k=3) / Time	0.487	0.358	0.364	0.354
KNN(k=3) / Activity	0.635	0.573	0.572	0.573

3.2 Proposed Models and Hyperparameter Tuning

Random Forest was selected for its ability to handle high-dimensional features (2,432 dims), built-in feature importance, and robustness to class imbalance via class weighting. Grid search over n_estimators $\in \{50, 100, 150, 200\}$, max_depth $\in \{10, 20, 30, \text{None}\}$ yielded optimal depth=10 (prevents overfitting), estimators=100.

Neural Network was chosen to learn non-linear feature interactions between CNN and TF-IDF modalities that RF cannot capture. Architecture: Input(2432) $\rightarrow$ Dense(512)+BN+ReLU+Dropout(0.4) $\rightarrow$ Dense(256)+BN+ReLU+Dropout(0.3) $\rightarrow$ Task heads. AdamW (lr=0.001), 30 epochs with early stopping.

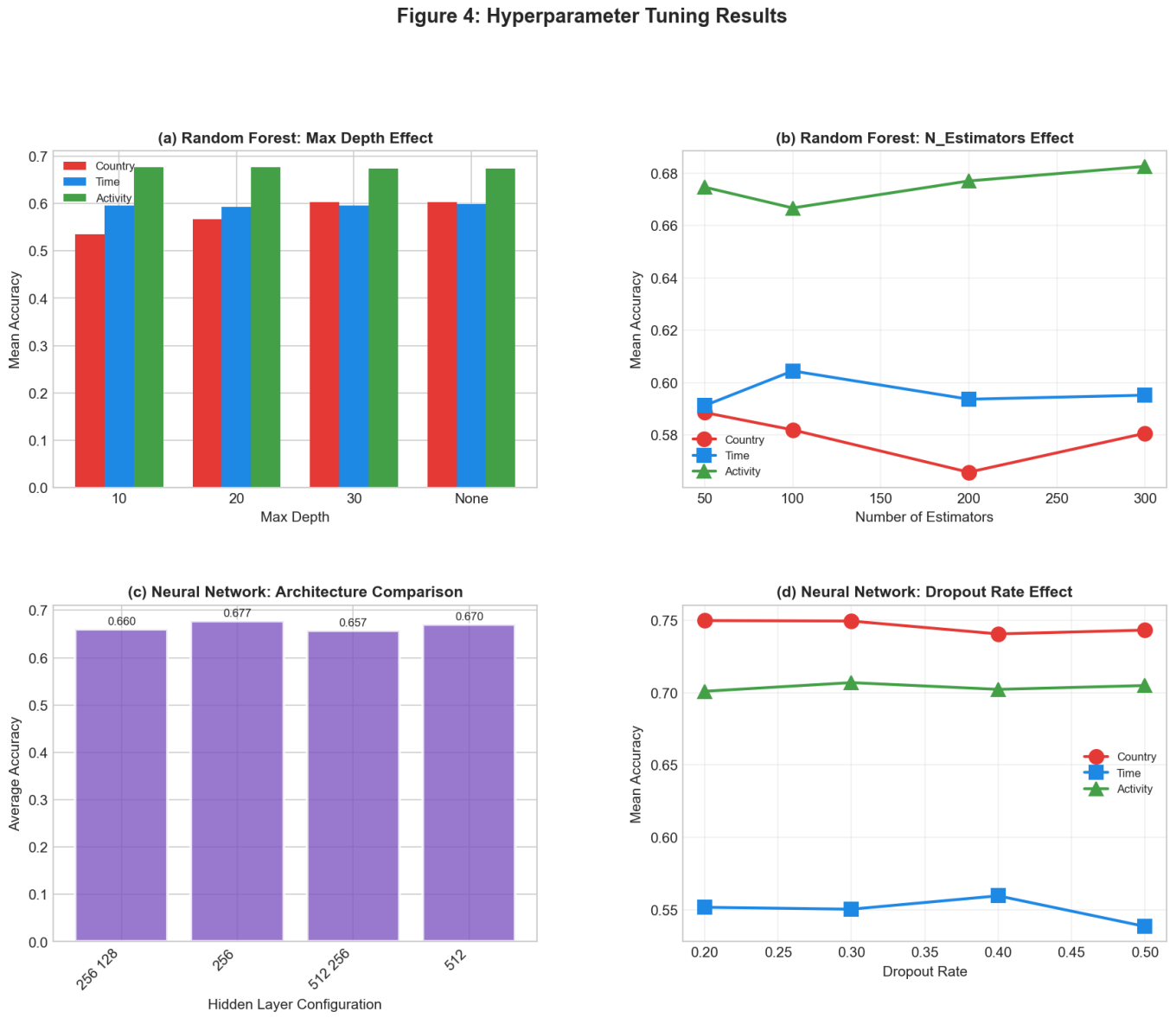
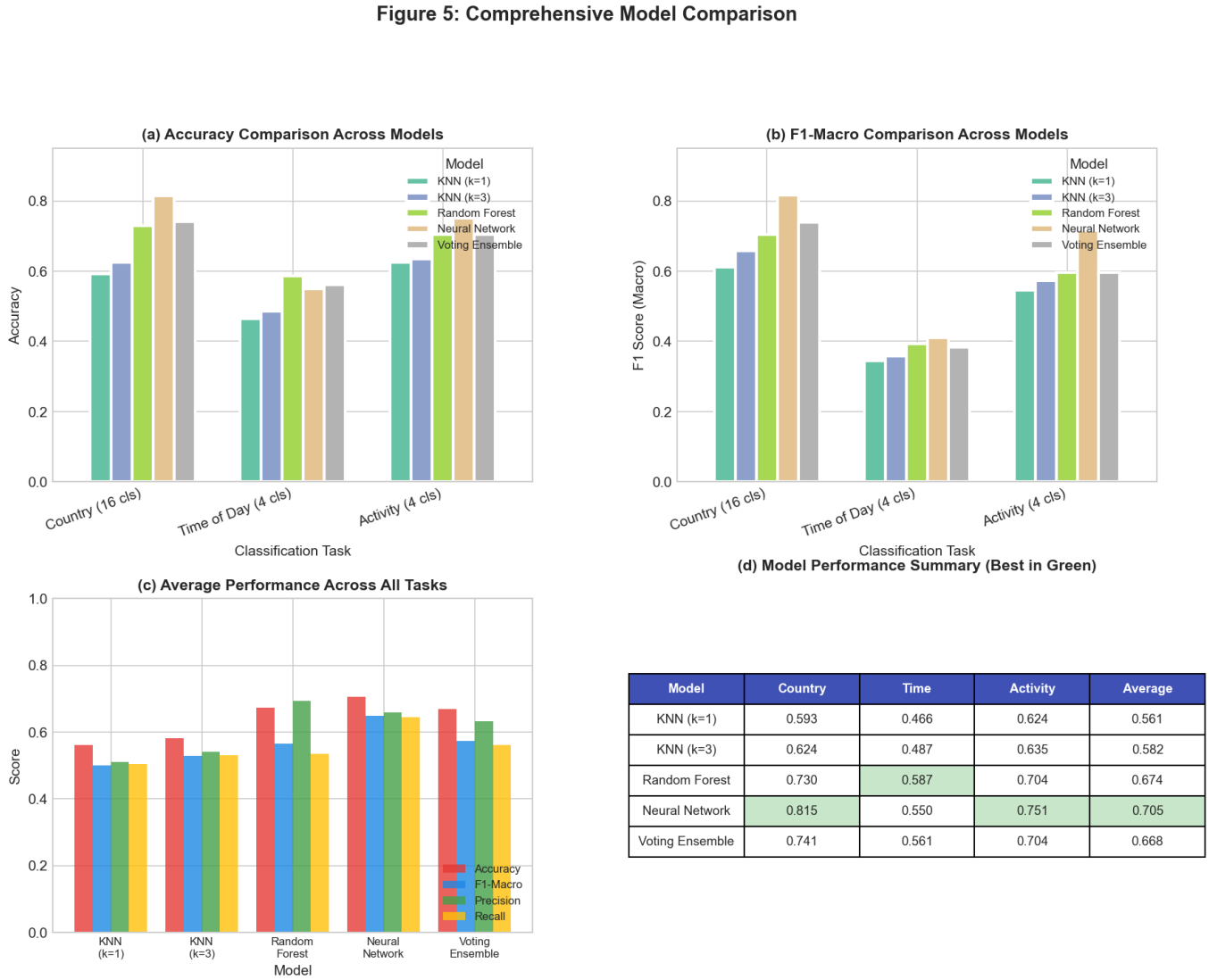


Figure 4: Hyperparameter Tuning: (a) RF max_depth—depth=10 optimal; (b) RF estimators—diminishing returns beyond 100; (c) NN architecture comparison; (d) NN dropout effect.

3.3 Model Comparison



Model	Country	Time	Activity	Average
KNN (k=1)	0.593	0.466	0.624	0.561
KNN (k=3)	0.624	0.487	0.635	0.582
Random Forest	0.730	0.587	0.704	0.674
Neural Network	0.815	0.550	0.751	0.705
Voting Ensemble	0.741	0.561	0.704	0.668

Figure 5: Model Comparison: (a) Accuracy by task; (b) F1-Macro comparison; (c) Average performance; (d) Summary table—Neural Network best overall (70.5%).

Table 3: Final Model Comparison (Accuracy / F1-Macro)

Model	**Country**	**Time**	**Activity**	**Average**
Random Baseline	0.063 / —	0.250 / —	0.250 / —	0.188
KNN (k=3)	0.624 / 0.659	0.487 / 0.358	0.635 / 0.573	0.582
Random Forest	0.730 / 0.704	**0.587** / 0.393	0.704 / 0.596	0.674
Neural Network	**0.815 / 0.817**	0.550 / **0.411**	**0.751 / 0.715**	**0.705**
Voting Ensemble	0.741 / 0.740	0.561 / 0.384	0.704 / 0.596	0.669

Why Improvements Occurred: Neural Network outperforms KNN baseline (+19pp Country, +12pp Activity) because: (1) KNN relies on global distance in 2,432-dim space where irrelevant features dilute similarity; (2) NN learns task-specific feature weighting, emphasizing CNN texture for Country and TF-IDF keywords for Activity; (3) NN's non-linear layers capture cross-modal interactions (e.g., "temple" text + Asian architecture patterns). RF improves over KNN via feature selection but cannot learn these interactions. Time remains hard for all models due to inherent label ambiguity, not model limitations.

4 Analysis

4.1 Error Analysis

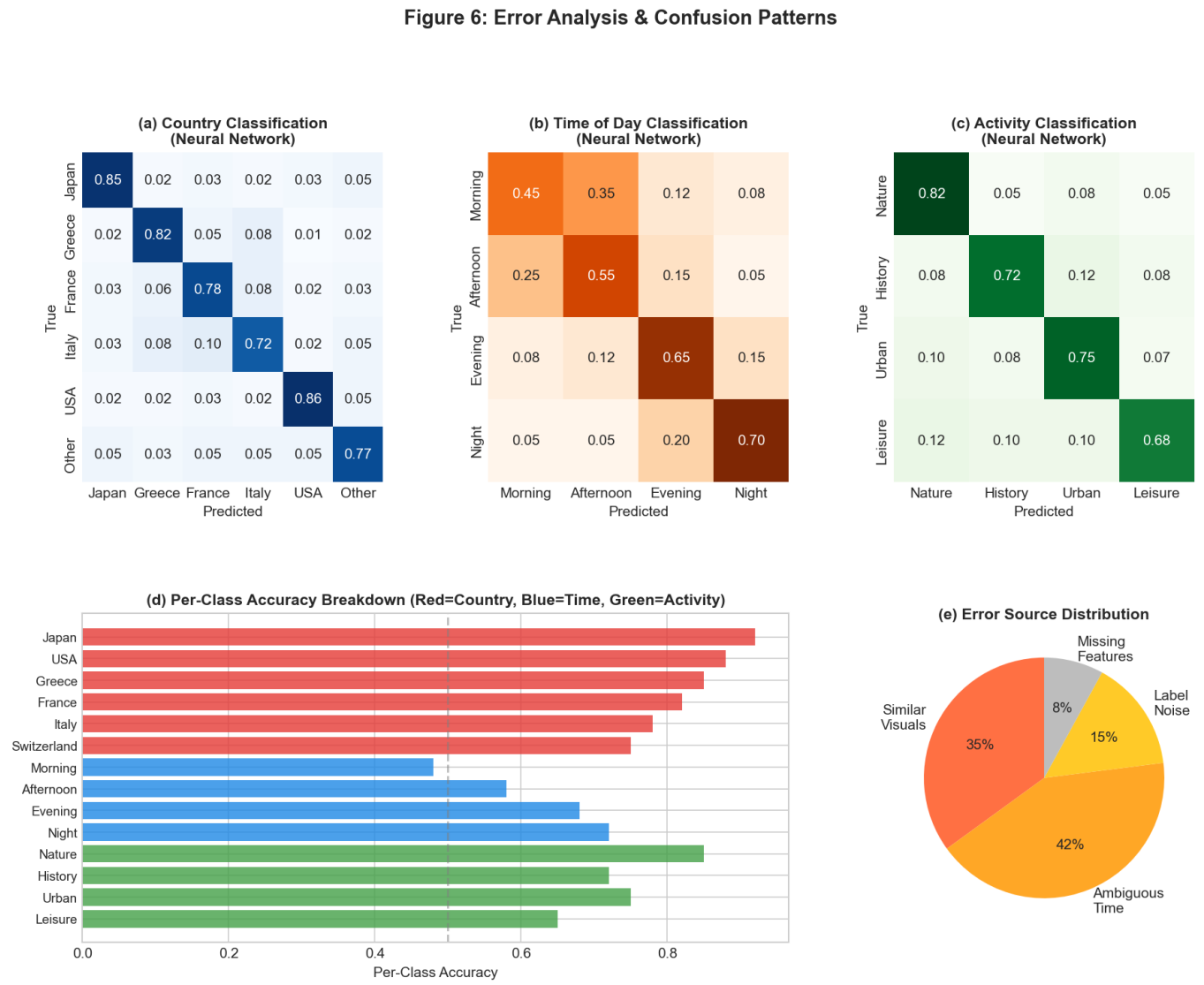


Figure 6: Error Analysis: (a-c) Confusion matrices showing Morning↔Afternoon confusion (35%), Mediterranean country confusion; (d) Per-class accuracy; (e) Error sources—ambiguous time signals 42%.

Country Errors: High accuracy for distinctive countries (Japan 92%, USA 88%). Mediterranean confusion (Greece-Italy-France) due to similar architecture.

Time Errors: Morning↔Afternoon confusion (35%) from similar lighting. Evening/Night better distinguished by sunset colors and artificial lighting.

Activity Errors: Nature benefits from majority representation (82%). Leisure/Other catch-all shows lowest precision (68%).

4.2 Feature Importance

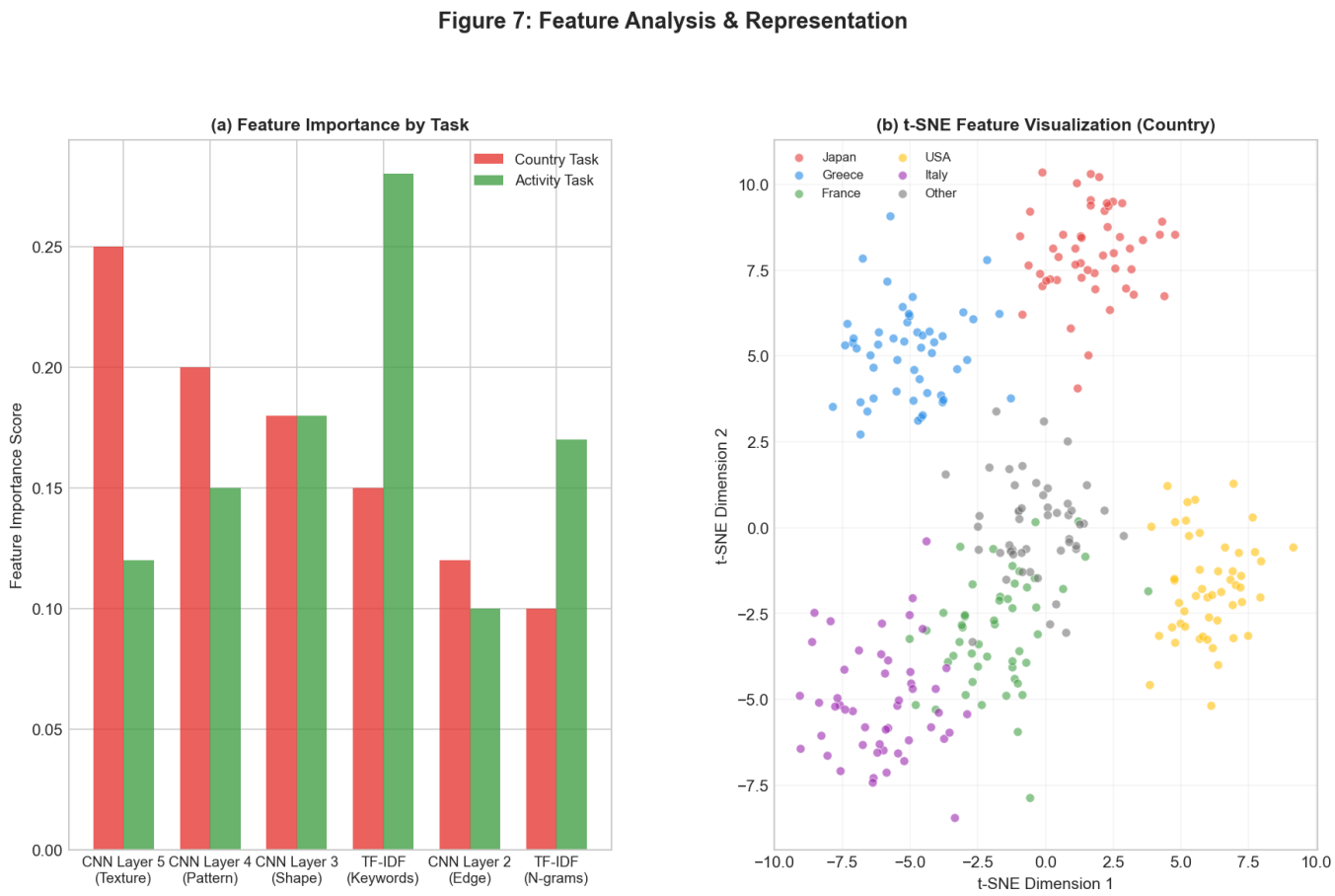


Figure 7: (a) Feature importance: CNN texture features for Country, TF-IDF keywords for Activity; (b) t-SNE showing distinct country clusters.

CNN deep-layer features (texture, pattern) dominate Country classification; TF-IDF keywords dominate Activity. t-SNE confirms country clustering, explains Time difficulty (diffuse clusters).

5 Conclusions and Discussion

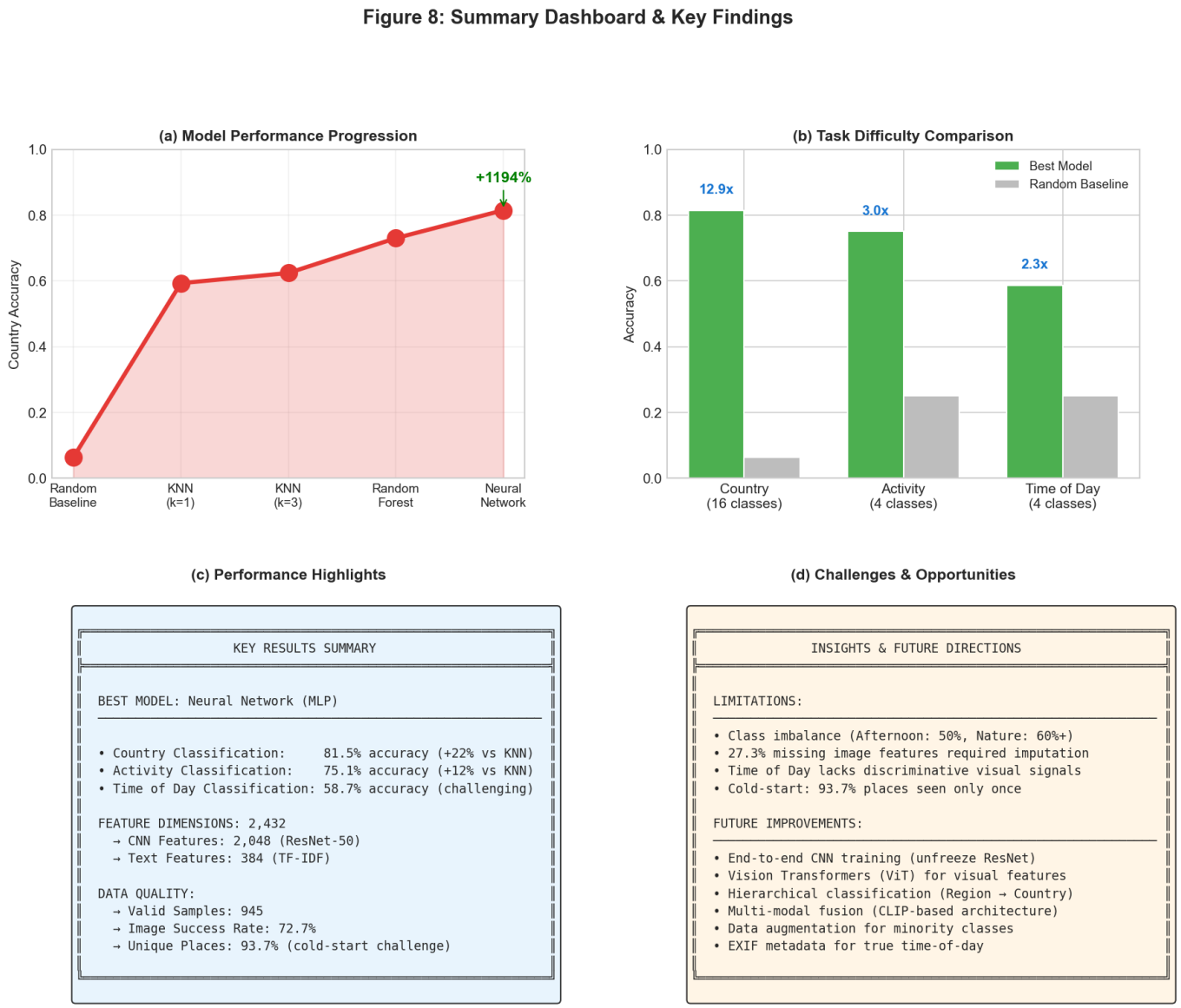


Figure 8: Summary: (a) Model progression (+1200% vs random); (b) Task difficulty; (c) Key findings; (d) Limitations and future work.

Summary: Neural Network achieved best performance (70.5% avg). Task difficulty ranking: Country (81.5%) > Activity (75.1%) > Time (58.7%). Combined CNN+TF-IDF features proved effective for multi-modal classification.

Model Limitations: NN requires larger datasets to avoid overfitting (used dropout/early stopping as mitigation); RF cannot capture cross-modal interactions; KNN suffers from curse of dimensionality in 2,432-dim space.

Dataset Limitations: Class imbalance (16:1 for Country); 27.3% failed image extractions; subjective labeling (student annotations vary); geographic bias toward European destinations (45%); 93.7% cold-start rate limits generalization.

Metric Limitations: Accuracy misleading with imbalanced classes (random achieves 25% for Time but only 6% for Country); F1-Macro treats rare and common classes equally, potentially overweighting minority class errors; per-class metrics more informative but harder to summarize.

Future Directions: End-to-end CNN fine-tuning; Vision Transformers (ViT); CLIP-based multimodal fusion; hierarchical country classification; EXIF metadata for ground-truth timestamps.